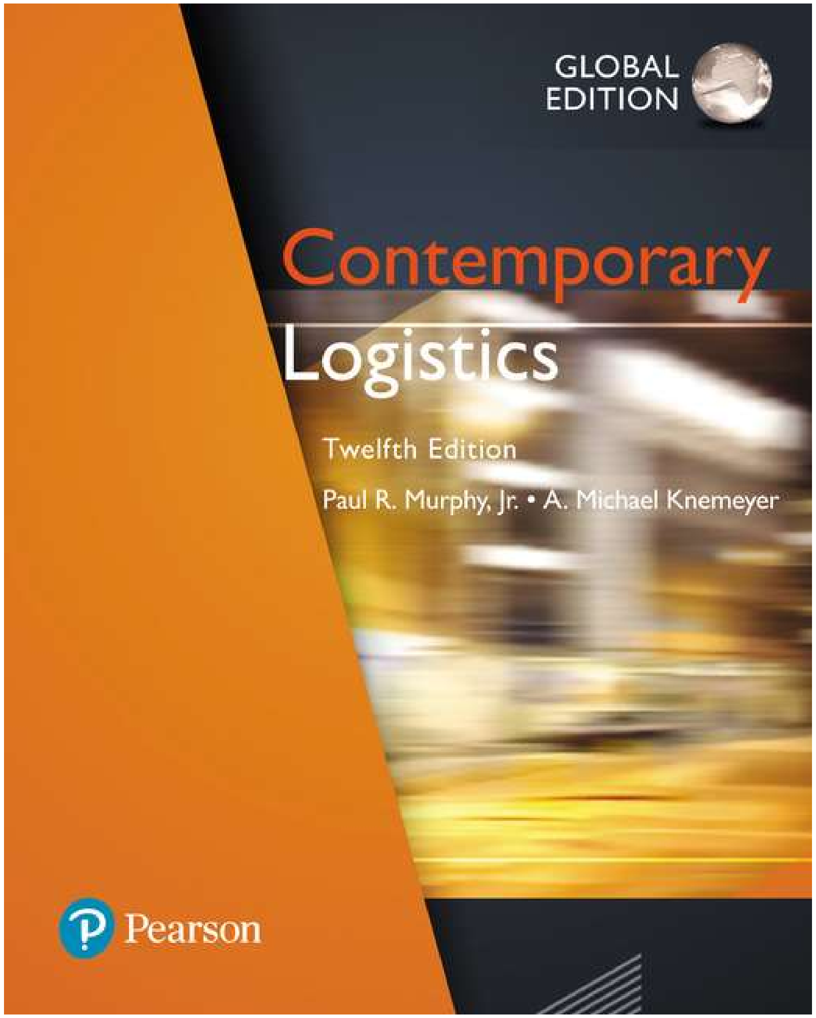
Contemporary Logistics

Twelfth Edition, Global Edition



Chapter 11

Packaging and Materials Handling

Why do you think packaging decisions affect transportation cost?

Learning Objectives

11.1 To illustrate how product characteristics affect packaging and materials handling

11.2 To discuss packaging fundamentals, such as packaging functions and labeling

11.3 To identify select issues that affect packaging, such as environmental protection and packaging inefficiencies

11.4 To learn about unit loads and the unit load platform

11.5 To explain materials handling, materials handling principles, and materials handling equipment

“Why do you think packaging decisions affect transportation cost?”

Product Characteristics (1 of 9)

- Physical characteristics
  - Substance form (solid, liquid, and gas)
  - Ability to withstand exposure to elements (rain, humidity)
  - Product density (weight per volume)
  - Product perishability
- Chemical characteristics
  - Incompatible products (sensitive to ethylene vs ethylene emitting)
- Characteristics must be made known to consumers (labeling)

Figure 11.1: Fabric Care Label

Source: Vartanov Anatoly/Shutterstock

Figure 11.2: Lumber Markings

Source: Alamy

Packaging Fundamentals (2 of 9)

- Packaging
  - Refers to materials used for the containment, protection, handling, delivery, and presentation of goods[1]
- Building-blocks concept
  - Products -> crates or boxes -> Pallets
  - A very small unit is placed into a slightly larger unit, and so on, to protect the product
  - Small unit packaging complements large unit packaging.

[1] *Logistics Dictionary*, www.tntfreight.com

Figure 11.5 Unit Loads *Source:* Dmitry Kalinovsky/Shutterstock

Packaging Fundamentals (3 of 9)

- Some of the many packaging fundamentals include:
  - Functional trade-offs (promote, protect, identify)
  - Packaging testing and monitoring
  - Labeling

Packaging Fundamentals (4 of 9)

- Functional trade-offs
- Packaging testing and monitoring
- Labeling

- Functional trade-offs
  - Packaging serves three general functions:
    1. To promote
    2. To protect
    3. To identify the relevant product
  - Packaging design decisions involve a number of departments within an organization

-Engineering	-Quality Control
-Manufacturing	-Transportation
-Marketing	-Warehousing

"If you design luxury packaging, what problem might logistics face?"

Packaging Fundamentals (5 of 9)

- Package testing and monitoring
  - Three important kinds of information needed to properly design protective packaging system:
    1. Severity of the distribution environment
    2. Fragility of the product to be protected
    3. Performance characteristics of various cushion materials (foams, air bulbs covers, thermocol)
  - Advisable to have packages pretested:
    - Vibrations, Dropping, Horizontal impacts, Compression, Rough handling (can be done by carriers)
  - Monitoring of packages facilitated by sensors

- Functional trade-offs
- Packaging testing and monitoring
- Labeling

Drop Testing – IPG Packaging Lab

Packaging is not just protection — it is a **risk management tool** in logistics.

Would you prefer cheaper packaging with higher risk, or expensive packaging with lower risk? Why?

Packaging Fundamentals (7 of 9)

- Functional trade-offs
- Packaging testing and monitoring
- Labeling

- Labeling:
  - Typically occurs at the end of the assembly process
  - Boxes must be labeled when contents are hidden
  - Many regulations govern labeling:
    - Weight, Specific contents, Instructions for use
  - Regulations differ from country to country and from state to state
  - Labeling comes in different forms:
    - Words, pictures, or codes (barcodes & batch number)

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Figure 11.3: Examples of Shipping Labels

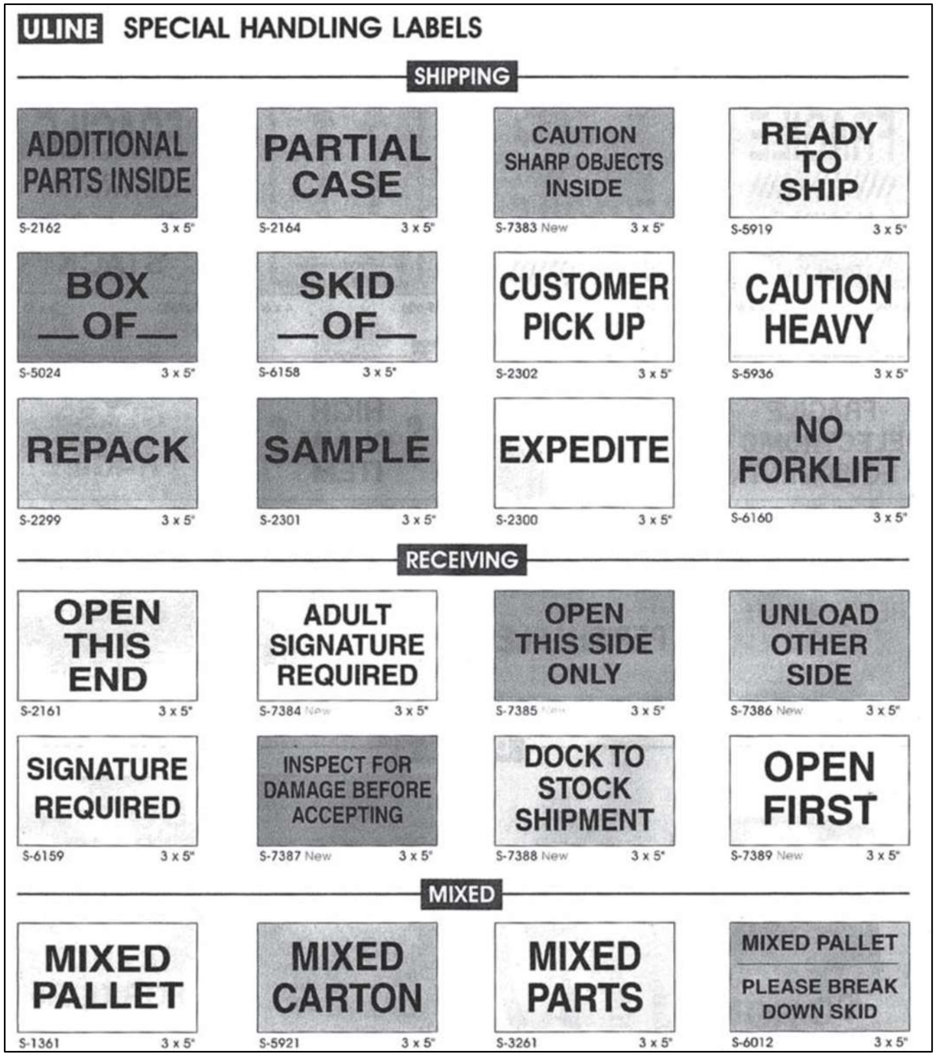


Figure 5.14 (a) *Bar code;* (b) *logistic label.*

Source: Courtesy of Uline. www.uline.com

Packaging Fundamentals (8 of 9)

- Hazardous materials
  - Governmental regulations address labeling of hazardous materials
  - Requirements involve:
    - Labeling
    - Packaging and repackaging
    - Placing warnings on shipping documents
    - Notifying transportation carriers in advance
    - Packages, containers, trailers, and railcars carrying hazardous materials must carry distinct placards identifying the hazard

Packaging Fundamentals (9 of 9)

- Hazardous materials
  - Globally Harmonized System of Classification and Labeling of Chemicals (GHS) is a global system to classify and label hazardous materials
  - GHS provides three key pieces of classification and labeling information:
    - A symbol
    - A signal word (e.g., “danger”)
    - A hazard statement (e.g., “explosion; severe projection hazard”)
  - Implemented by fewer than 75 countries to date

Issues in Packaging (1 of 4)

Packaging and global waste

- Packaging materials (plastics, papers, glass, aluminum) accounts for 28% of municipal solid waste (MWS)[1]
- Environmental protection
  - Reduce packing materials used
  - Use packaging materials that are more environmentally friendly with recycled content (Biodegradable)
  - Use reusable containers (closed-loop system)
  - Retain or support services that collect used packaging and recycle it (closed-loop system)
- Initiative such as zero packaging materials such as Pieter Pot in the Netherlands is disrupting the supermarket chains

1 www.epa.gov

Issues in Packaging (2 of 4)

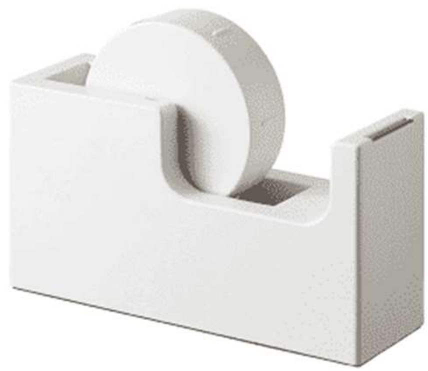
- Metric system
  - U.S., Liberia, and Myanmar (formerly Burma) are the only three countries in the world that do not use the metric system of measurement
  - Increasing pressure on U.S. exporters to market their products overseas in metric units

Issues in Packaging (3 of 4)

- Identifying packaging inefficiencies
  - Building-blocks concept is useful for analyzing packaging inefficiencies
  - Packaging inefficiencies can have a number of undesirable logistics consequences including:
    - Increased loss
    - Increased damage
    - Slower materials handling
    - Higher storage costs
    - Higher transportation costs

Table 11.1: A Hypothetical Example of Packaging Inefficiency

Product:	One (1) desktop tape dispenser (cube = 30 inches) 12 dispensers per carton
Product cube per carton:	30 cubic inches times 12 dispensers = **360 cubic inches**
Carton dimensions:	1,140 cubic inches
Carton efficiency:	Product cube per carton divided by carton cube **360/1,140 = 31.6%**
60 cartons can be put on a pallet	
Pallet capacity:	90,720 cubic inches
Carton cube per pallet:	1,140 cubic inches per carton times 60 cartons = **68,400 cubic inches**
Pallet load efficiency:	Carton cube per pallet divided by pallet capacity **68,400/90,720 = 75.4%**
Carton efficiency times pallet load efficiency = the amount of pallet cube that is actually product:	**31.6% times 75.4% = 23.8%**



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Issues in Packaging (4 of 4)

- Packaging's influence on transportation considerations
  - Carriers' tariffs and classifications influence the type of packaging and packing methods that must be used
    - Fedex and UPS switched to calculating shipping rates based on dimensional weight (space by weight)
  - Carriers established classifications for two main reasons:
    - Packaging specifications determined by product density lead to the best use of the equipment's weight and volume capabilities
    - Carrier specifications for protective packaging reduce likelihood of damage to products, thus reducing the loss and damage claims filed against the carrier

Figure 11.4: Boxmaker's Guarantee

YouTube video

Source: © Pearson Education.

Unit Loads in Materials Handling (1 of 5)

- Packaging of materials is based on building-blocks concept of putting products in containers to provide efficient and manageable units.
- Unit load (unitization)
  - Refers to consolidation of several units (cartons or cases) into larger units to improve efficiency in handling and to reduce shipping costs[2]
- Handling efficiency
  - Facilitated by mechanical devices (pallet jack or forklift) as well as by using a pallet or skid

[2] *Source:* http://cscmp.org/digital/glossary/glossary.asp

Pallet handling systems

Move horizontally

Pallet jack

Pedestrian electric trans-pallet

stand-on electric trans-pallet

(a) (b) (c)

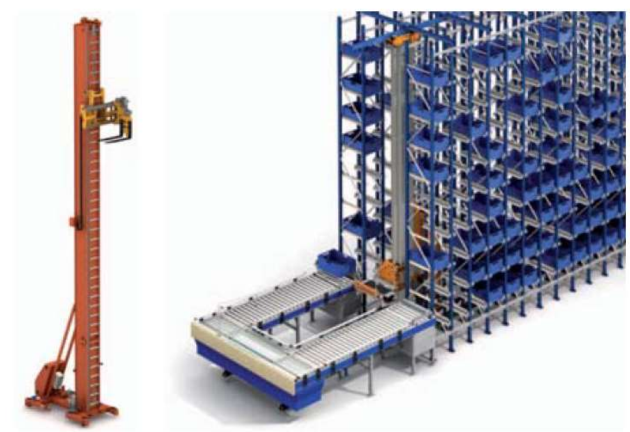
ıre 5.8 *(a) Counterbalanced forklift truck; (b) reach truck; (c) vertical order*

Automated guided vehicle. Reproduced by permissi

Move horizontally & vertically

Unit Loads in Materials Handling (2 of 5)

- Advantages
  - Additional protection to cargo (boxes are secured to pallet with wrappings and straps)
  - Pilferage is discouraged
  - More valuable or fragile items can be stacked inside the unit load to provide them with additional protections
  - Mechanical devices can substitute for manual labor ([Example](#))



(a) (b)

(a) Stacker crane for pallets; (b) AS/RS system serve

Unit Loads in Materials Handling Limitations (3 of 5)

- Provides large quantity that sometimes is of limited value to resellers dealing in smaller quantities
- Must use costly mechanical or automated device to move
- Lack of standard pallet size

Dimensions Millimeters (width × length)	Dimensions Inches (width × length)
1219 × 1016	48.00 × 40.00
1000 × 1200	39.37 × 47.24
1165 × 1165	44.88 × 44.88
1067 × 1067	42.00 × 42.00
1100 × 1100	43.30 × 43.30
800 × 1200	31.50 × 47.24

Most widespread:

1. Europallet 800x1000 ISO1
2. Anglo-Saxon pallet 1000x1200 ISO2
3. Australian pallet 1050x1050

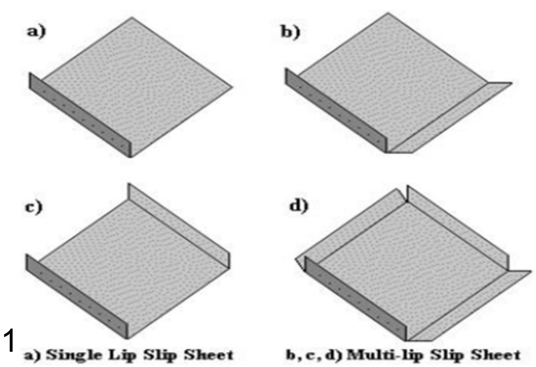
Table 5.2 Box size (in cm) compatible with filling the surface of the europallet and the Anglo-Saxon pallet.

	y	$y/2$	$y/3$	$y/4$	$y/5$
x	60 × 40	60 × 20	60 × 13.3	60 × 10	60 × 8
$x/2$	30 × 40	30 × 20	30 × 13.3	30 × 10	30 × 8
$x/3$	20 × 40	20 × 20	20 × 13.3	20 × 10	20 × 8
$x/4$	15 × 40	15 × 20	15 × 13.3	15 × 10	15 × 8
$x/5$	12 × 40	12 × 20	12 × 13.3	12 × 10	12 × 8

Unit Loads in Materials Handling (4 of 5)

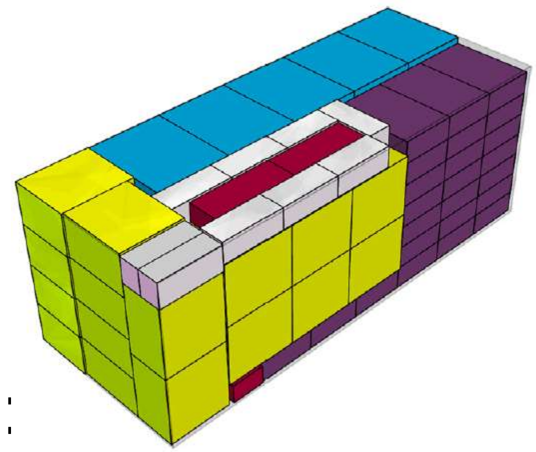
- Basic unit is a pallet
  – Can be constructed from wood (90%), plastic, wood composites (such as fiberboard), paper, and metal
  – Each pallet material has advantages and disadvantages
    1. Price is a major drawback of plastic and metal pallets
    2. Longevity is advantage of plastic & metal (up to years life span)
    3. Plastic and metal are more resistance thus more safe
    4. Wood and plastic are inflammable
  – Pallet types have different weights, e.g, plastic 30 pounds, wood 40 pounds, metal 65pounds (100 pounds~45KG)
- Pallet or skid alternatives:

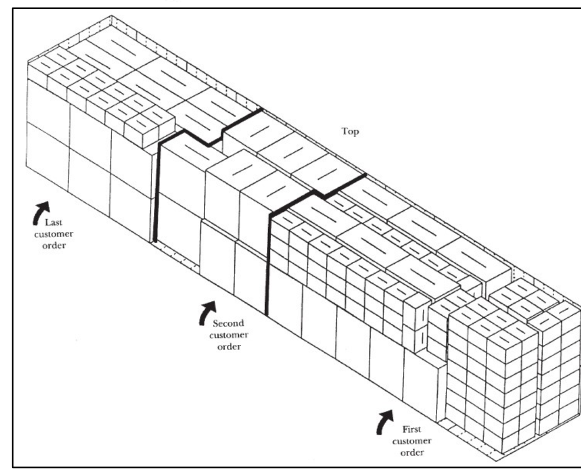
  Slip sheet



Unit Loads in Materials Handling (5 of 5)

- Beyond the unit load
  – Stowage based upon load-planning software (Bin packing problem)
  – Bracing or inflatable dunnage bags
  – Load is subjected various forces including:
    - Vibration
    - Pitch
    - Roll





- Weighing out: heavy good utilizes the railcar's, trailer's, or container's weight capacity without filling its cubic capacity

Materials Handling (1 of 6)

- Materials handling
  - Refers to the "short-distance movement that usually takes place within the confines of a building such as a plant or DC and between a building and a transportation service provider"[3]
  - Short-distance movement distinguishes materials handling from transportation
  - Products are handled by the building-block concept or large quantities are handled in bulk (eg, coal, grain, and iron ore)
  - Handling may change the characteristics (or quality) of the product

[3] John J. Coyle, C. John Langley, Jr., Brian J. Gibson, Robert A. Novack, and Edward J. Bardi, *Supply Chain Management: A Logistics Perspective*, 8th ed. (Mason, OH: South-Western Cengage Learning, 2009), Appendix 11-A.

Figure 11.8: Conveyor Belt for Bulk Materials

Source: Shutterstock

Materials Handling Principles (2 of 6)

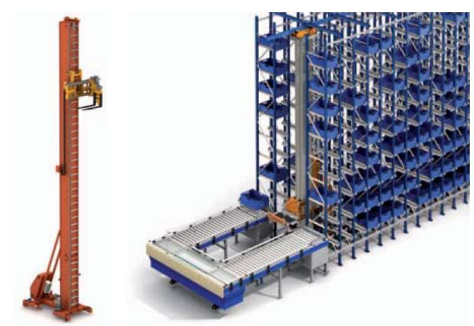
Materials handling principles are important for warehousing design and performance improvement:

1. *Planning* focusing proactive rather than on reactive
2. *Standardization* focuses on materials handling activities, processes, and equipment
3. W*ork principle* emphasizes working smarter, rather than harder
4. *Ergonomics* refers to the science that seeks to adapt work or working conditions to suit the abilities of the worker.
5. *Unit load principle* focuses on maximizing the value of unit loads.
6. S*pace utilization principle* strives to make the best use of existing space
7. S*ystem principle* emphasizes that the materials handling function impacts other logistics functions and activities.
8. A*utomation principle* suggests that, where appropriate, automation can positively impact efficiency and reliability
9. *Environmenta*l impact and energy consumption considers as a criteria when designing or selecting alternative equipment and material handling systems.
10. *Life cycle cost principle is* thorough economic analysis should account for the entire life cycle of all material handling equipment and resulting systems.

Materials Handling Equipment (3 of 6)

- Materials handling equipment can be divided into
  - Storage equipment
    - Shelves
    - Racks
    - Bins
  - Handling equipment
    - Conveyor systems
    - Lift trucks
    - Carts
    - Cranes

Figure 5.11 (a) Fixed roller conveyor; (b) fixed wheel conveyor. Reproduced by permission of OMT BIELLA.

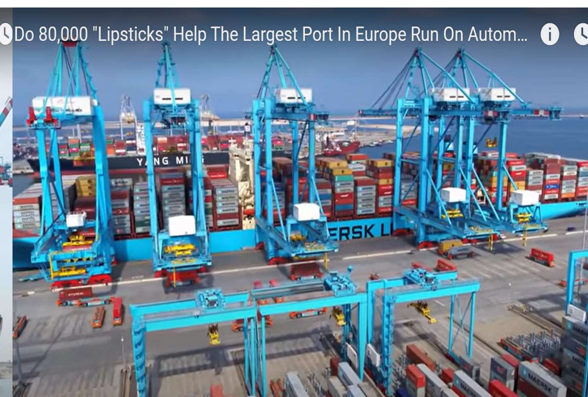


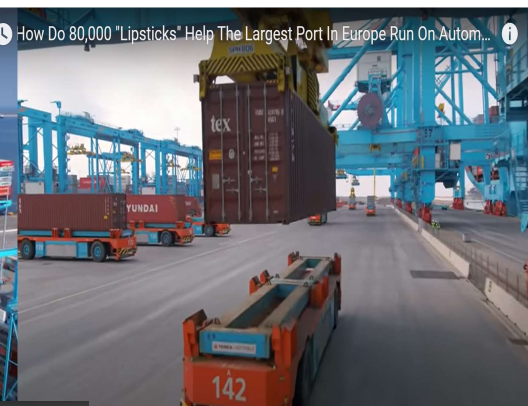
(a) (b)

(a) Stacker crane for pallets; (b) AS/RS system serve

Materials Handling Equipment (4 of 6)

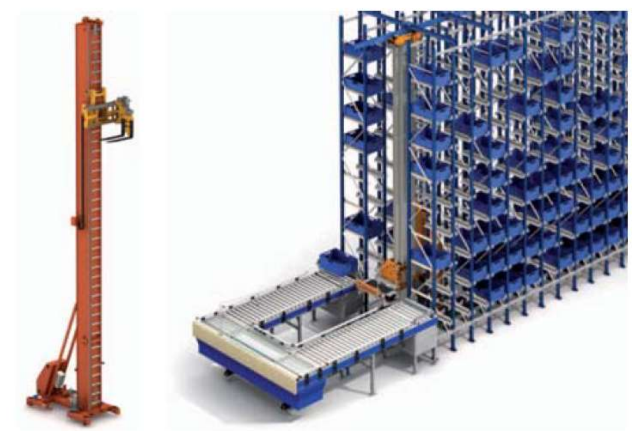
- Can also be categorized in terms of:
  - Labor intensive, Mechanized (e.g., forklifts), Automated (e.g., AGVs)
- Sufficient volume is needed to justify high cost of automated equipment





Materials Handling Equipment (5 of 6)

– The choice of handling equipment can influence the type of storage equipment
– The choice of storage equipment can influence the type of handling equipment
– An organization's order picking and assembly system can also influence the type of handling equipment
  - Picker-to-part systems (Forklift, AR)
  - Part-to-picker systems (Carousel)



(a) (b)

(a) Stacker crane for pallets; (b) AS/RS system serve

Key Terms (1 of 2)

- Building-blocks concept
- Closed-loop systems
- Cubes out
- Dimensional weight (dim weight)
- Ergonomics
- Globally Harmonized System of Classification and Labeling of Chemicals (GHS)
- Materials handling (material handling)
- Packaging
- Pallet (skid)

Key Terms (2 of 2)

- Part-to-picker system
- Picker-to-part system
- Slip sheet
- Unit load (unitization)
- Weighing out